Reg. No. | | | | | | | | | | | [12EC73]

SEVENTH SEMESTER B.E EXAMINATION DECEMBER –2015

OPTICAL FIBER COMMUNICATION

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	**a.**	Derive an expression for the numerical operture of a step index fiber (SIF) from Snell's law.	**06**
	b.	Explain photonic crystal fibers.	**04**
	c.	Explain disadvantages of Optical fiber communication.	**04**
2.	**a.**	Explain intermodal (Chromatic) dispersion and polarization mode dispersion.	**05**
	b.	Briefly explain signal attenuation.	**03**
	c.	A 30Km long fiber at 1300mm has an attenuation of 0.8db/Km. If 200μw power is launched into the fiber; find the o/p power in dBm and in watts.	**06**
3.	**a.**	Explain and derive the expressions for quantum efficiency and LED power.	**08**
	b.	A double hetero junction in GaASP LED emitting at a peak wave length of 1310 mm has radiative and non-radiative recombination times of 30 and 100ms, respectively. The drive current is 40mA calculate the **i.** Bulk recombination Life time **ii.** Internal quantum efficiency **iii.** Internal power level.	**06**
4.	**a.**	Explain the following **i.** Any two types of splicing **ii.** Fibre end face preparation	**08**
	b.	An optical source having refractive index of 3.6 coupled to fiber of 1.8 refractive index. Consider the medium between fiber and source has similar index as that of fiber. Calculate the Fresnel reflection and loss of power in dB.	**06**
5.	**a.**	Explain general; configuration of an eye diagrams showing the definition of fundamental measurements parameter. And also explain noise margin and timing jitter parameter.	**07**
	b.	Explain coherent detection. Write the difference between coherent and non-coherent detection.	**07**

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6. a. Explain the link power budget, with a relevant diagram. **06**

b. Explain the following: **08**

i. Subcarrier Multiplexing.

ii. Mode-Partition noise.

7. a. Explain the operation of polarization independent isolator made of three miniature optical components. **06**

b. Explain the importance of the following active components used in WDM based on MEMS: **08**

i. Variable optical attenuators

ii. Tuneable optical filters.

8. a. What are applications of optical amplifiers? **03**

b. Write short note semiconductor optical amplifiers. **04**

c. Explain with relevant diagram SONET/SDH rings. **07**